MEDICATION SAFETY SIGNAL MONITOR · HAND-IN R00

Start here

Version 1.1, 6 November 2026: what this is, what to read first, and what the words mean

What this is	What this is not
• The front door to the hand-in set • One glossary, one code key, one evidence legend	• A results report (the evidence is in R01 to R08) • Clinical advice: a signal is a lead, not proof of harm

How to read this. Read sections 1 and 2 first. Keep sections 3 to 5 open as look-up tables while you read the other reports. Section 6 and the dark box are the facts to remember.

1 · What this project is

	In one line
The user and her questions	**Dana**, a hospital pharmacy safety officer (fictional persona, R04 ch. 2), asks each morning: (1) Which drugs have side effects reported that the label does not mention? (2) Which have an active recall? (3) What has changed since yesterday?
The trust rule	Every flag traces to its source: the label versions searched (DailyMed) and the FDA recall record open in one click; the FAERS case reports are reachable by query, not yet from her screen.
The 13 drugs	**8 opioids:** oxycodone, hydrocodone, morphine, fentanyl, hydromorphone, methadone, buprenorphine, tramadol. **5 GLP-1 drugs** (diabetes and weight loss): semaglutide, liraglutide, dulaglutide, tirzepatide, exenatide. Exenatide has no reference label, so signals exist for 12 of 13 (D018, D031 v5).
Data and platform	Public data only, no patient records. Built on **Palantir Foundry** with its AI layer **AIP**. The app is one Workshop module of 4 pages (v0.10.0).

2 · The hand-in set, in the order to read it

#	Read	It answers
1	Architecture note: picture and build facts	What is the whole system? (Skip its decision boxes at first.)
2	R07 Data judgment, chapter 1	What would messy public data do to the officer's morning?
3	R04 The application, chapters 1–4	How does the officer use the app, and what does each number mean?
4	R01 Foundry from scratch	What is Foundry, which parts did we use, and where do I click?
5	R02 Pipelines: raw to ontology	How does raw public data become signals, recalls and changes?
6	R03 The Ontology (ch. 6 = NAION trace)	How does a flag link back to its source?
7	R06 Optimization · R08 AI reliability · R07 rest	What did we measure? How often is the AI wrong? Which data problems did we fix or leave?
8	R05 Decisions, and what we did not do	Why each choice, and what we left out. Read last, as the index.
+	Deck and R09 Demo navigation guide	What the 20-minute demo shows, click by click, and what each stop proves.

Files: `docs/11-handin/architecture-note/architecture-note.pdf` · `docs/11-handin/deck/MSSM-demo-deck.pdf` · `docs/11-handin/reports/pdf/R01…R09`. Order from `docs/11-handin/reviews/first-time-reader.md` §6.

3 · Glossary

Term	Plain meaning (Foundry and general)	Term	Plain meaning (drug safety)
Palantir Foundry	A data platform holding data, the code that cleans it, a model of real things and the apps, with version history.	**FAERS**	FDA Adverse Event Reporting System: voluntary reports of suspected side effects, quarterly. Moved to **AEMS** in March 2026 (DQ20).
Dataset	A versioned table in Foundry. Ours run raw → clean → curated.	**Case**	One FAERS report of one patient event; only its latest version counts (D022).
Transform	Python code that reads datasets and writes one. One run = a *build*.	**Suspect drug**	A drug the reporter suspects (primary, secondary, interacting); drugs only *also taken* do not count (D021).
Code Repository	A git repository in Foundry holding the transforms.	**MedDRA PT**	Preferred term: the standard word for a reaction. Its groupings need a licence we lack.
Data Lineage	The graph of which dataset feeds which, with build status.	**2×2 table**	For a drug and reaction: *a* = both; *b* = drug, other reactions; *c* = other drugs, this reaction; *d* = the rest.
Ontology	Foundry's layer of objects, links and Actions on curated datasets.	**PRR**	Proportional reporting ratio = (a ÷ the drug's reports) ÷ (c ÷ other drugs' reports): how much more often the reaction comes with this drug.
Object (type)	One real thing / its kind (Drug); one dataset row each.	**Evans rule**	A common screen: 3+ reports, PRR ≥ 2, chi-squared ≥ 4 (D006).
Link type	A named relationship between two object types, e.g. Signal → Drug.	**DailyMed / SPL**	The official US label site / its XML format (Structured Product Labeling). Our label truth (D019).
Action	A form that writes to the Ontology: ours saves a Triage Decision (*write-back*).	**Label section**	A titled part of a label. We search Boxed Warning, Contraindications, Warnings (and Precautions), Adverse Reactions.
Workshop	Foundry's point-and-click app builder; Dana's screen is one.	**openFDA**	The FDA's public web API: recall, Drugs@FDA and NDC data.
AIP	Palantir's AI Platform: language models in pipelines. Ours reads label sections.	**Recall**	FDA enforcement report of a product withdrawn or corrected; Class I most serious. *Active* = Ongoing.
F1	One test score balancing *found every right item* (recall) and *added no wrong ones* (precision). 1.0 is perfect.	**RxNorm / RxCUI**	US standard drug vocabulary / its concept id. Our drug key = ingredient RxCUI, resolved once (D020).
NAION	Sudden sight loss from optic-nerve damage; semaglutide × NAION is our worked example. It is a *known positive*, not a discovery: EMA PRAC (6 Jun 2025) and a WHO alert (27 Jun 2025) named it; our 3 US labels do not mention it (DECISIONS learning 2026-11-05).	**NDC**	National Drug Code: the US code for one package.
		Reference label	The label versions searched for a drug (Drug → Products → label).

4 · Key to the ID codes

Code	Means	Look it up in
D###	A decision (where the brief is silent). `D016 v3` = third version; old ones kept. The owner accepted the last 12 open ones in one batch on 6 Nov 2026 (D006, D015, D025–D029, D037, D045–D048), plus D041 v2 (the AI also reads the semaglutide labels).	`docs/06-decisions/DECISIONS.md` (D001–D048)
A###	An approach we tried, with its verdict: kept, dropped or parked.	`docs/05-approaches/INDEX.md` (A000–A019)
DQ##	A data-quality issue found in the public data.	`docs/07-data-quality/data-issues-register.md` (DQ01–DQ43)
O#	An optimization, measured before and after (O1, O1b, O2, O3, O3a).	`docs/08-optimization-log/optimization-log.md`
G1–G6	Gate defaults in the dataset contracts, accepted via D039.	`docs/04-architecture/dataset-contracts-v1.md`
Step N	12-step build order; each step ends with a proof (*gate*). 6–8 = first slice; 9 = all 13 drugs; 10 = incremental FAERS; 11 = screen + health checks; 12 = hand-in.	`docs/04-architecture/foundry-conventions/project-build-order-v1.md`

Code	Means	Look it up in
Baseline gate	The full rebuild where every hand-in number was frozen.	`docs/11-handin/baseline-gate.md`
R##	A hand-in report, R00 (this page) to R09.	`docs/11-handin/reports/pdf/`
30–38	Diagrams: 30 data flow (v3), 31 ontology (v2), 32 signal pipeline (v3), 33 incremental (v3), 34 change feed, 35 monitoring (v2), 36 demo map (v3), 37 ripple map (v3), 38 one-page architecture (v2). Older versions are kept.	`docs/diagrams/README-handin.md`

5 · Evidence labels and status bands

Each claim carries an evidence label (what kind of claim); each chapter-closing box a status band (how far along). Text read live from `tools/handin/style.py`, so it matches every report.

Evidence	Meaning	Status	Meaning
OBSERVED	Read directly from a public source (FAERS, DailyMed, openFDA, RxNorm), with the query that produced it.	TESTED	Built, run, and checked against evidence.
DERIVED	Computed by our pipeline from observed data, for example a PRR or a count per drug and reaction.	PLANNED	On the plan; not built yet.
ASSUMED	Not verified: taken as true for now because the data or the brief is silent. It could be wrong.	PROPOSED	Suggested; waiting for a decision.
DESIGNED	A design choice we made and wrote down (the decision log says why). It may be built; the status band says how far.		

6 · Five numbers and what they mean

Number	What it means	Do not read it as	
1,952 signals	Drug–reaction pairs that meet the Evans rule **and** whose reaction is not found in any reference label of the drug (1,844 *not in label*) or only via a medium-band synonym (108 *possibly in label*).	Proven side effects. It means more reports than expected, not cause.	DERIVED
1,073 ranked	The **review queue**: signals *not in label*, in the clinical bucket (D016 v3), under half from mass reports of 30+ drugs (D023 v2). 1,355 at step 9; synonyms and buckets cut it.	1,073 findings.	DERIVED
126 active recalls	openFDA recalls with status **Ongoing**, out of 900 recalls pulled for our ingredients.	*Not terminated* (that would add Completed).	OBSERVED
26 changes	Change events (signal, recall, label) in the **last 30 days** (Morning brief card). Page 3's date range gives *since my last review* (D045).	Changes since yesterday.	DERIVED
F1 0.97	AI label reading on a 35-item answer key (Boxed Warning + Contraindications of 2 oxycodone labels; written by Claude, frozen before the run, no pharmacist yet); found 35 of 35 items; target 0.85 (D015). Separately **745/745** extracted terms are grounded (quote found) over all 21 sections of 5 labels (2 oxycodone, 3 semaglutide).	A score for all 745 terms, or for the semaglutide labels (not scored).	DERIVED

What this means for Dana (the safety officer)

FAERS is about 130 days behind. Signals change only when a new FAERS quarter arrives (here 2026Q1-2026Q2), so *what changed* is mostly recalls and labels.

FAERS moved to AEMS in March 2026 (DQ20). Where future quarters will come from is not verified yet.

Only the project owner can submit a Triage Decision today (R03 ch. 5); giving Dana that right is a production step.

The AI's quotes are on her screen (page 1, *What the AI read in these labels*, Workshop v0.10.0; R04 ch. 7).

Three checks agree NAION is not in the label. Exact text and synonyms found nothing in the 3 semaglutide labels, and the AI, which read all 3, found diabetic-retinopathy terms but no optic-nerve term.

The Morning brief *changes* card is the last 30 days, not since yesterday. Use page 3's date range.

Gate: How we know this page is right TESTED

Required evidence: All 26 numbers are `docs/11-handin/numbers.json` lookups (captured 2026-11-05T07:50Z; keys listed in this page's source file); definitions copied from the Phase D fix brief; each code names a real file.

Result: Met: builds with no layout problems; every page looked at.

If it fails: Fix this page first: the other reports rely on its meanings.